SUB-MODULE 7.2 · PARAMETER-EFFICIENT FINE-TUNING

Prefix Tuning

Trainable soft prompts prepended to every Transformer layer's key-value space — no weight modification

9 min	B3 – Apply	L3 – Advanced	AI-FT-IN-TH-000008
Duration	Bloom's	Level	ID

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Prefix Tuning — Core Idea

Definition (Li & Liang, ACL 2021)

Prefix Tuning prepends a sequence of trainable continuous vectors (the 'prefix') to the key-value pairs at every Transformer layer. Unlike hard prompts (discrete tokens), prefix vectors are continuous and optimised directly via gradient descent.

- The prefix is not constrained to the discrete token vocabulary — it can represent any direction in activation space
- Parameterised through a reparameterisation MLP during training for stability: prefix = MLP(prefix_params). The MLP is discarded after training — only the prefix embeddings are stored
- Number of trainable parameters: $L \times P \times 2 \times d$ (L = layers, P = prefix length, d = model dimension)
- Fully separable from the base model: the base weights are never touched

Prefix Tuning — Architecture Diagram

Input Tokens
$x_1 \dots x_n$

Prefix Vectors
P_K, P_V (TRAINED)

Multi-Head Attention (Frozen)

Keys = concat(P_K, W_K·x)

Values = concat(P_V, W_V·x)

Queries = W_Q·x (unchanged)

Attention = softmax(Q·K^T/√d)·V

This same prefix is applied at every Transformer layer — P_K and P_V are layer-specific. The prefix acts as a soft, task-specific 'memory' that conditions the model's attention at every layer simultaneously. This is more powerful than a single soft prompt at the input only (prompt tuning).

Prefix Tuning vs Hard Prompts vs Soft Prompt Tuning

✅ Advantages over Hard Prompts

- Continuous optimisation — not constrained to real tokens
- Can represent any concept, not just expressible vocabulary
- End-to-end optimised for the task — no manual prompt engineering
- Gradient signal is informative — no straight-through estimator needed

⚠️ Disadvantages vs LoRA

- Consumes context window: P prefix tokens used per call at inference
- Generally underperforms LoRA on tasks requiring deep parameter adaptation
- Training can be unstable without the reparameterisation MLP trick
- Prefix length is a hyperparameter requiring tuning (typically 10–100 tokens)

Prefix Tuning Hyperparameters

Parameter	Typical Range	Effect
Prefix length P	10–100 tokens	Longer prefix = more capacity but more context consumed at inference
Reparameterisation MLP hidden dim	512–2048	Larger MLP = more stable training; discarded post-training
Prefix dropout	0.0–0.1	Regularisation; prevents overfitting to training distribution
Applied layers	All transformer layers (standard)	Applying to all layers is key; input-only prefix is much weaker
Task format	Constrained generation (table→text, summarisation)	Best performance on structured output tasks

KEY REFERENCES

- Li, X.L. & Liang, P. (2021). Prefix-Tuning: Optimizing Continuous Prompts for Generation. ACL 2021.
- Lester, B. et al. (2021). The Power of Scale for Parameter-Efficient Prompt Tuning. EMNLP 2021.
- Liu, X. et al. (2022). P-Tuning v2: Prompt Tuning Can Be Comparable to Fine-Tuning. arXiv.

▶ **Next: Concept 10: Adapter Modules (AI-FT-IN-TH-000009)**